

Worksheet

Computer Network - Week 3 - DNS

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Activity 1 : nslookup

Screenshots

```
C:\Windows\system32>nslookup -type=NS iitb.ac.in
Server:   UnKnown
Address:  10.6.47.254

Non-authoritative answer:
iitb.ac.in      nameserver = dns3.iitb.ac.in
iitb.ac.in      nameserver = dns2.iitb.ac.in
iitb.ac.in      nameserver = dns1.iitb.ac.in

iitb.ac.in      nameserver = dns2.iitb.ac.in
iitb.ac.in      nameserver = dns1.iitb.ac.in
iitb.ac.in      nameserver = dns3.iitb.ac.in
dns3.iitb.ac.in internet address = 103.21.127.129
dns2.iitb.ac.in internet address = 103.21.126.129
dns1.iitb.ac.in internet address = 103.21.125.129
dns2.iitb.ac.in internet address = 103.21.126.129
dns1.iitb.ac.in internet address = 103.21.125.129
dns3.iitb.ac.in internet address = 103.21.127.129
```

Answers

1. What is the IP address of www.iitb.ac.in?

10.6.47.254

2. What is the IP address of the DNS server that provided the answer to your nslookup command in

question 1 above?

103.21.125.129

3. Was the response from your nslookup command in question 1 authoritative or non-authoritative?

It is non-authorative

4. Use nslookup to determine the authoritative name server for the domain www.iitb.ac.in. What is its name? (If there is more than one authoritative server, what is the name of the first authoritative server returned by nslookup?) If you need to find the IP address of that authoritative name server, how would you do it?

dns1.iitb.ac.in, in my command line, it is in the bottom line of the output, there are the name server with its IP address

Activity 2 : Flushing DNS Cache

Screenshots

```
C:\Windows\system32>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::91e7:9651:c820:5862%4
    IPv4 Address. . . . . : 10.6.47.43
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 10.6.47.254

Tunnel adapter isatap.{DE912EB7-21B5-46EB-B7AE-01CD9A8A5485}:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Tunnel adapter Teredo Tunneling Pseudo-Interface:

    Connection-specific DNS Suffix  . : 
    IPv6 Address. . . . . : 2001:0:2851:782c:1068:38ef:f5f9:d0d4
    Link-local IPv6 Address . . . . . : fe80::1068:38ef:f5f9:d0d4%2
    Default Gateway . . . . . : ::
```

Ipconfig to check my IP address

Wireshark packet capture showing a DNS query for writing.engr.psu.edu. The packet list shows a standard query (0x63d) from 10.6.47.43 to 10.6.47.254. The packet details pane shows the query structure, including the transaction ID (0x63d) and the query name (writing.engr.psu.edu). The packet bytes pane shows the raw data in hexadecimal and ASCII.

Frame 44: 80 bytes on wire (640 bits), 80 bytes captured (640 bits) on interface \Device\NPF_{DE912EB7-21B5-46EB-B7AE-01C9A8A5485}, Id 0

Ethernet II, Src: EliteGro_30:2d:36 (94:c6:91:30:2d:36), Dst: 9e:e3:74:3e:cc:10 (9e:e3:74:3e:cc:10)

Internet Protocol Version 4, Src: 10.6.47.43, Dst: 10.6.47.254

User Datagram Protocol, Src Port: 64022, Dst Port: 53

Domain Name System (query)

Transaction ID: 0x63d

Flags: 0x0100 Standard query

Questions: 1

Answer RRs: 0

Authority RRs: 0

Additional RRs: 0

Queries

writing.engr.psu.edu: type A, class IN

Name: writing.engr.psu.edu

[Name Length: 20]

[Label Count: 4]

Type: A (Host Address) (1)

Class: IN (0x0001)

[Response Tr: 81]

Request message

Wireshark packet capture showing a DNS response for writing.engr.psu.edu. The packet list shows a standard query response (0x63d) from 10.6.47.254 to 10.6.47.43. The packet details pane shows the response structure, including the transaction ID (0x63d) and the response name (writing.engr.psu.edu). The packet bytes pane shows the raw data in hexadecimal and ASCII.

Frame 81: 266 bytes on wire (2128 bits), 266 bytes captured (2128 bits) on interface \Device\NPF_{DE912EB7-21B5-46EB-B7AE-01C9A8A5485}, Id 0

Ethernet II, Src: 9e:e3:74:3e:cc:10 (9e:e3:74:3e:cc:10), Dst: EliteGro_30:2d:36 (94:c6:91:30:2d:36)

Internet Protocol Version 4, Src: 10.6.47.254, Dst: 10.6.47.43

User Datagram Protocol, Src Port: 53, Dst Port: 64022

Domain Name System (response)

Transaction ID: 0x63d

Flags: 0x8100 Standard query response, No error

Questions: 1

Answer RRs: 2

Authority RRs: 4

Additional RRs: 4

Queries

writing.engr.psu.edu: type A, class IN

Name: writing.engr.psu.edu

[Name Length: 20]

[Label Count: 4]

Type: A (Host Address) (1)

Class: IN (0x0001)

Answers

writing.engr.psu.edu: type CNAME, class IN, cname coe-a10-01.ucts.psu.edu

Name: writing.engr.psu.edu

Type: CNAME (Canonical Name for an alias) (5)

Class: IN (0x0001)

Response message

Answers

1. To which IP address was the DNS request sent? Is this the IP address of your default local DNS server?

Yes, my ip address = 10.6.47.43

2. Check the DNS request message. What is the “Type” of this DNS request? Does the request message contain an “Answer”?

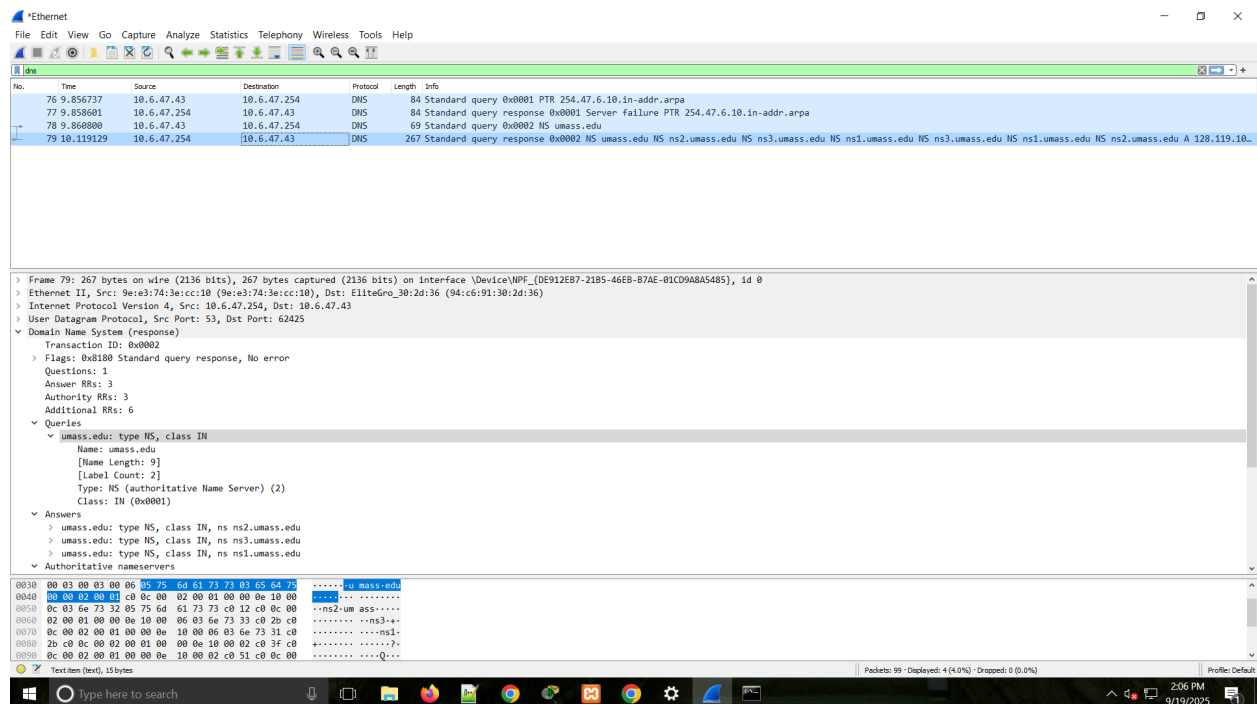
Type = A (Host Address), no it doesn't contains the “answer”

3. Check the DNS response message for that request. How many “Questions” are included in the DNS response? How many “Answers”?

Questions : 1
Answer RRs : 2

Activity 3 : Capture & DNS Message Analysis with Wireshark

Screenshots



Answers

1. To which IP address was the DNS request sent? Is this the IP address of your default local DNS server?

Yes, my ip address = 10.6.47.43

2. Check the DNS request message. How many questions does the request contain? Does the request message include an "Answer"?

It has one "question", yes it contain the "answer"

3. Check the DNS response message. How many answers does the response contain? What information is included in the answers? How many additional resource records are returned? What additional information is included in these additional resource records?

It has three answer; it contains the authoritative name server with its detail information like the name, type, class ; it returned six additional resource record; same like in "answer" but with its IP address